Let $u = \cos^2 x,\ du = -\sin(2x)\,dx,$ $\qquad$ $\begin{array}{c|c|c} x & 0 & \frac{\pi}{4} \\ \hline u & 1 & \frac{1}{2} \end{array}$

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} \sin(2x)\ln(\cos x)\,dx &= \frac{1}{2}\int_{\frac{1}{2}}^{1} \ln u\,du \\ &= \frac{1}{2}[u\ln u - u]_{\frac{1}{2}}^{1} \\ &= \frac{1}{4}(\ln 2 - 1)\end{aligned}$$

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} \cos(2x)\ln(\cos x)\,dx &= \left[\frac{1}{2}\sin(2x)\ln(\cos x)\right]_0^{\frac{1}{4}\pi} + \frac{1}{2}\int_0^{\frac{1}{4}\pi} \sin(2x)\tan x\,dx \\ &= -\frac{1}{4}\ln 2 + \frac{1}{2}\int_0^{\frac{1}{4}\pi} 2\sin^2 x\,dx \\ &= -\frac{1}{4}\ln 2 + \frac{1}{2}\int_0^{\frac{1}{4}\pi} (1-\cos 2x)\,dx \\ &= -\frac{1}{4}\ln 2 + \frac{1}{2}\left[x - \frac{1}{2}\sin 2x\right]_0^{\frac{1}{4}\pi} \\ &= \frac{1}{8}(\pi - \ln 4 - 2)\end{aligned}$$

Let $t = x - \dfrac{\pi}{4},\ dx = dt,$ $\qquad$ $\begin{array}{c|c|c} x & \frac{\pi}{4} & \frac{\pi}{2} \\ \hline t & 0 & \frac{\pi}{4} \end{array}$

$$\cos x + \sin x = \sqrt{2}\cos t$$

$$\cos(2x) + \sin(2x) = \cos 2t - \sin 2t$$

$$\begin{aligned}\int_{\frac{1}{4}\pi}^{\frac{1}{2}\pi} (\cos(2x)+\sin(2x))\ln(\cos x+\sin x)\,dx &= \int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\ln(\sqrt{2}\cos t)\,dt \\ &= \int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\left(\frac{1}{2}\ln 2 + \ln(\cos t)\right)\,dt \\ &= \frac{1}{2}\ln 2\int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\,dt + \int_0^{\frac{1}{4}\pi} \cos 2t\ln(\cos t)\,dt - \int_0^{\frac{1}{4}\pi} \sin 2t\ln(\cos t)\,dt\end{aligned}$$

$$\int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\,dt = \left[\frac{1}{2}\sin 2t + \frac{1}{2}\cos 2t\right]_0^{\frac{1}{4}\pi} = 0$$

$$\int_{\frac{1}{4}\pi}^{\frac{1}{2}\pi} (\cos(2x) + \sin(2x)) \ln(\cos x + \sin x)\, dx$$
$$= \frac{1}{2} \ln 2 \cdot 0 + \int_0^{\frac{1}{4}\pi} \cos 2t \ln(\cos t)\, dt - \int_0^{\frac{1}{4}\pi} \sin 2t \ln(\cos t)\, dt$$
$$= \frac{\pi}{8} - \frac{1}{2} \ln 2$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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